

Bhuwan Sharma

Computational & Systems Biology

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RESEARCH INTERESTS

Graph neural networks for biological systems • biological network analysis • biomarker discovery • protein dynamics and flexibility • physics-informed machine learning • knowledge graph and Gene Ontology embeddings

EDUCATION

University of North Bengal

2023–2025

M.Sc. Bioinformatics

CGPA 7.65. Dissertation research conducted in the Computational Systems Biology Lab.

Kalimpong College, University of North Bengal

2020–2023

B.Sc. Zoology (Honours)

CGPA 7.64. Coursework included molecular biology, genetics, and biochemistry.

RESEARCH EXPERIENCE

Research Scholar

2025–Present

Computational Systems Biology Lab, Department of Bioinformatics, University of North Bengal

- Developing deep learning methods for cancer biomarker identification and protein conformational dynamics prediction.
- Designing graph neural network architectures that integrate protein–protein interaction networks, Gene Ontology embeddings, and knowledge graph representations.
- Building physics-informed graph models, incorporating quantum-inspired neural PDE operators, for protein dynamics.

SOFTWARE & TOOLS

CABiGoWEB <https://godmbc.compbiosysnbu.in/>

Web server implementing graph-based deep learning methods for cancer biomarker identification. It combines protein–protein interaction networks with Gene Ontology information and knowledge graph embeddings, and provides biomarker prediction and functional enrichment.

LGNM <https://lgnm.compbiosysnbu.in/>

Learnable Graph Network Model for protein conformational dynamics. The model learns heterogeneous residue interactions from molecular dynamics data using a physics-constrained graph formulation to predict residue-level flexibility.

ScriptaDocX <https://bhuwanshar.github.io/ScriptaDocX/>

Client-side document utility suite for PDF compression, OCR, merging, splitting, and watermarking.

METHODS

Graph Learning	Graph neural networks; GCN; GAT/GATv2; GraphSAGE; message passing; graph representation learning; biological network learning.
Biological Modelling	Protein–protein interaction networks; Gene Ontology; knowledge graphs; biomarker prioritization; differential expression; pathway and functional enrichment; multi-omics analysis.
Computational Biophysics	Elastic network models; protein flexibility; normal modes; molecular dynamics analysis; physics-informed graph models; residue-level interaction modelling.
Statistical Modelling	PCA; PCR; linear and multivariate regression; correlation and partial correlation; high-dimensional data analysis; network-based inference.

COMPUTATIONAL STACK

Programming	Python; R; Bash; C; Java; Perl.
ML Frameworks	PyTorch; PyTorch Geometric; DGL; scikit-learn; HuggingFace Transformers.
Scientific Computing	NumPy; SciPy; Pandas; CUDA; GPU computing; SQLite.
Bioinformatics	Bioconductor; limma; edgeR; DESeq2; clusterProfiler; biomaRt; ComBat; STRING; RDKit.
Networks	NetworkX; Cytoscape; Neo4j.

Infrastructure

Linux; Git/GitHub; Docker; Conda.

PUBLICATION

Sharma, B., & Sarkar, C. (2026). Learnable Graph Network Model (LGNM): A Physics-Constrained Graph Neural Network with Quantum Hamiltonian Learning. *bioRxiv*.